

Course Outline

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I. INTRODUCTION

The State of the World and Human Civilization: Year 2027

1. We Live in the Anthropocene
2. State of the Earth: Year 2027
 - (a) The Atmosphere
 - (b) The Hydrosphere
 - (c) The Cryosphere
 - (d) The Biosphere
 - (e) The Lithosphere
3. State of the Humans: Year 2027
 - (a) Current Societal Organizing Principles
 - (b) Current Technology
 - (c) Current Population
4. State of Human Civilization: Year 2027
 - (a) Globalized
 - (b) Weaponized
 - (c) Distracted
 - (d) Fragile

Opportunities for Planetary Stewardship

1. Stewardship as an Existential Imperative for Sustainability
2. The Critical Earth Systems Coupled to Human Welfare
 - (a) Global Climate System
 - (b) Terrestrial Ecosystems
 - (c) Aquatic Ecosystems
3. The Critical Production and Delivery Systems Coupled to Human Welfare
 - (a) Systems for Existential Human Needs
 - (b) Systems for Higher Human Needs
 - (c) Systems for Human Innovation
4. External Opportunities for Stewardship of Critical Systems
 - (a) Re-vamping Existing Systems
 - (b) Creating New Systems
5. Internal Opportunities for Stewardship of Critical Systems
 - (a) Developing Creativity
 - (b) Developing Intuition
 - (c) Caring for Mind, Body, and Spirit

II. APPROACHES

Systems Analysis

1. Fundamentals of Systems
 - (a) Definitions
 - (b) Systems Thinking
 - (c) Systems Science
2. Fundamentals of Complexity
 - (a) Definitions
 - (b) Complex Adaptive Systems
 - (c) Wicked Problems

3. Fundamentals of Scale

- (a) Spatial Scales
- (b) Temporal Scales

Data Analytics and Predictive Models

- 1. Probability and Statistical Models
- 2. Agent-Based Models
- 3. Imaginative Scenario Models

III. TOPICS

Standard Concepts and Frameworks of Sustainability

- 1. Standard Concepts of Sustainability
 - (a) Brundtland Commission
 - (b) UN Sustainable Development Goals
 - (c) Marshall and Toffel
- 2. Standard Frameworks for Sustainability
 - (a) Ecological Sustainability
 - (b) Social Sustainability
 - (c) Economic Sustainability
 - (d) Agricultural Sustainability
- 3. Standard Concepts of Resilience
 - (a) Robustness
 - (b) Rebounding Capacity
 - (c) Adaptability

Toward a Sustainable and Resilient Biosphere

- 1. Stewardship of the Global Climate System
 - (a) Necessary and Sufficient Conditions
 - (b) Best-Case Scenarios

2. Stewardship of Terrestrial Ecosystems
 - (a) Necessary and Sufficient Conditions
 - (b) Best-Case Scenarios
3. Stewardship of Aquatic Ecosystems
 - (a) Necessary and Sufficient Conditions
 - (b) Best-Case Scenarios

Toward a Sustainable and Resilient Civilization

1. Stewardship of Systems for Existential Human Needs
 - (a) Necessary and Sufficient Conditions
 - (b) Best-Case Scenarios
2. Stewardship of Systems for Higher Human Needs
 - (a) Necessary and Sufficient Conditions
 - (b) Best-Case Scenarios
3. Stewardship of Systems for Human Innovation
 - (a) Necessary and Sufficient Conditions
 - (b) Best-Case Scenarios